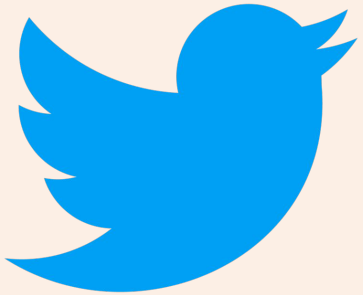
A Machine Learning Approach to

Detecting Twitter Bot Accounts



Developed by

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In association with

San Jose State University - CS 166 01 - Information Security, Dr. Chao-Li Tarng

Introduction

Twitter bots pose a threat to the security and integrity of online information, often serving as sources of misinformation and scams. Our project aims to develop a machine learning-based solution to accurately identify Twitter bots by analyzing behavioral patterns and account features, enhancing trust and safety on the platform.

Our Approach



Data Collection

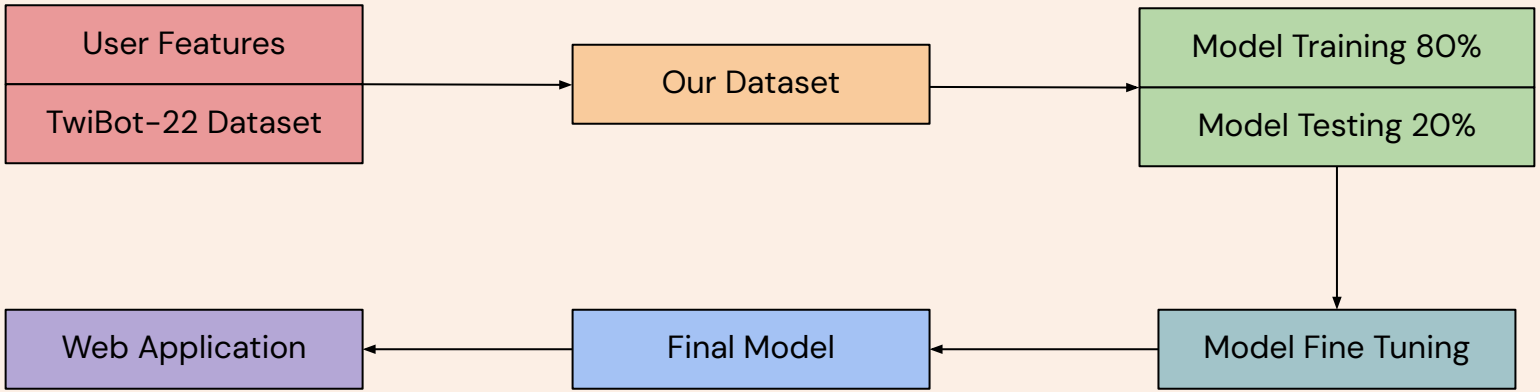


Model Creation



Web Application

Architecture & Design - Workflow



Architecture & Design - Datasets

TwiBot-22 (2022)

Benchmark Dataset

1,000,000 users

~14% bot

id	label
u15211869	human
u2465283662	bot
u1428341642	bot

Our Dataset

3,000 users

~50% bot

user_id	is_bot	avg_sentiment	identical_tweets_ratio	is_profile_image_valid
15211869	0	-0.191	0.0	1
2465283662	1	0.128	0.052	0
1428341642	1	0.367	0.155	1

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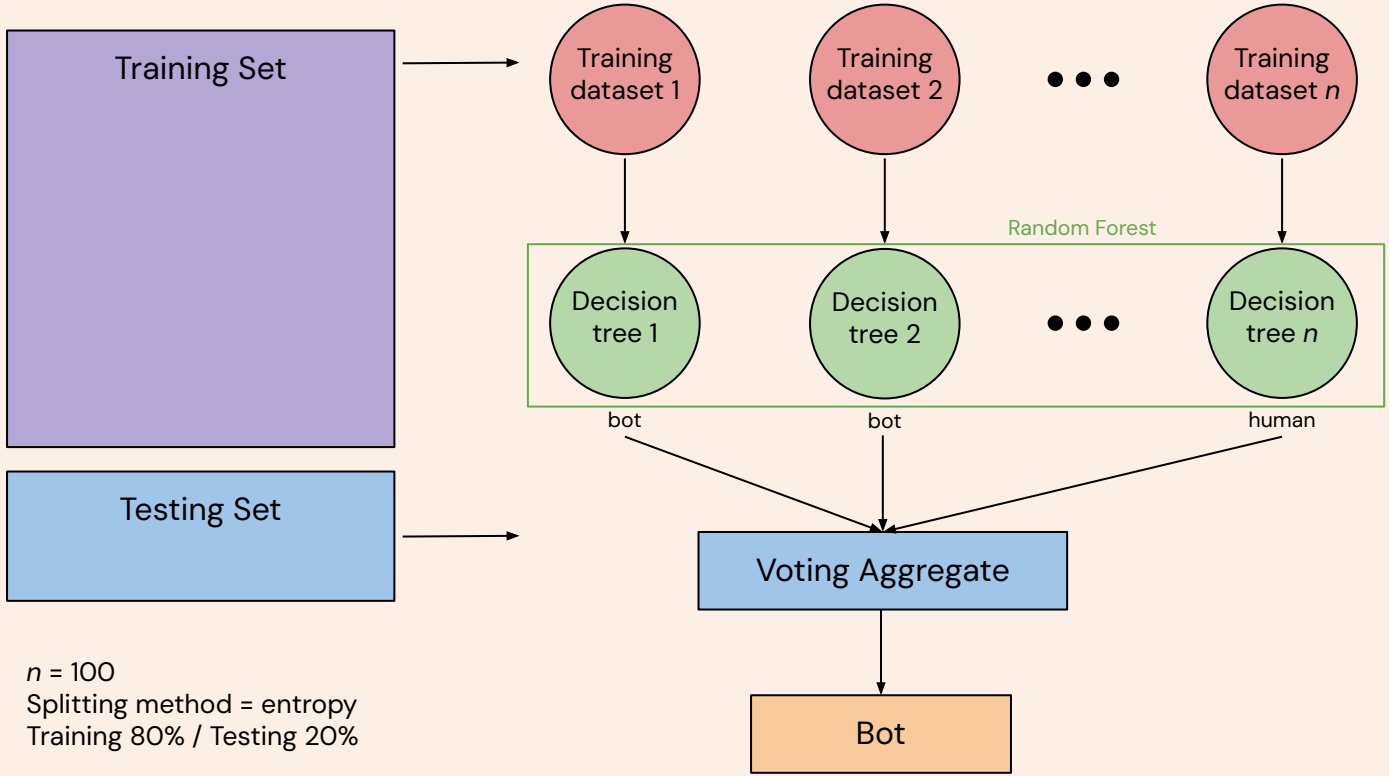
28 more user features

Architecture & Design - User Features

user_id **	followers_count	parsed_owned_tweets_count **	quotes_to_owned	avg_retweets_per_follower
screen_name **	following_to_followers	parsed_owned_text_tweets_count **	retweets_to_owned	
Is_bot **	is_possibly_sensitive	parsed_retweets_count **	avg_urls	
account_age	is_default_profile_image	likes_freq	avg_hashtags	
is_blue_verified	is_profile_banner	media_freq	identical_tweet_freq	
Is_verified **	is_profile_image_valid	followers_freq	avg_tweet_sentiment	
profile_description_sentiment	is_profile_banner	following_freq	avg_replies_per_follower	
following_count	tweet_freq	replies_to_owned	avg_likes_per_follower	

User Information User Activity Tweet Analysis Tweet Engagement

ML Algorithm - Random Forest



Bootstrap Sampling of a users' data for each set

Collection of decision trees from random samples

Take the voting majority from each of the trees

n = 100
Splitting method = entropy
Training 80% / Testing 20%

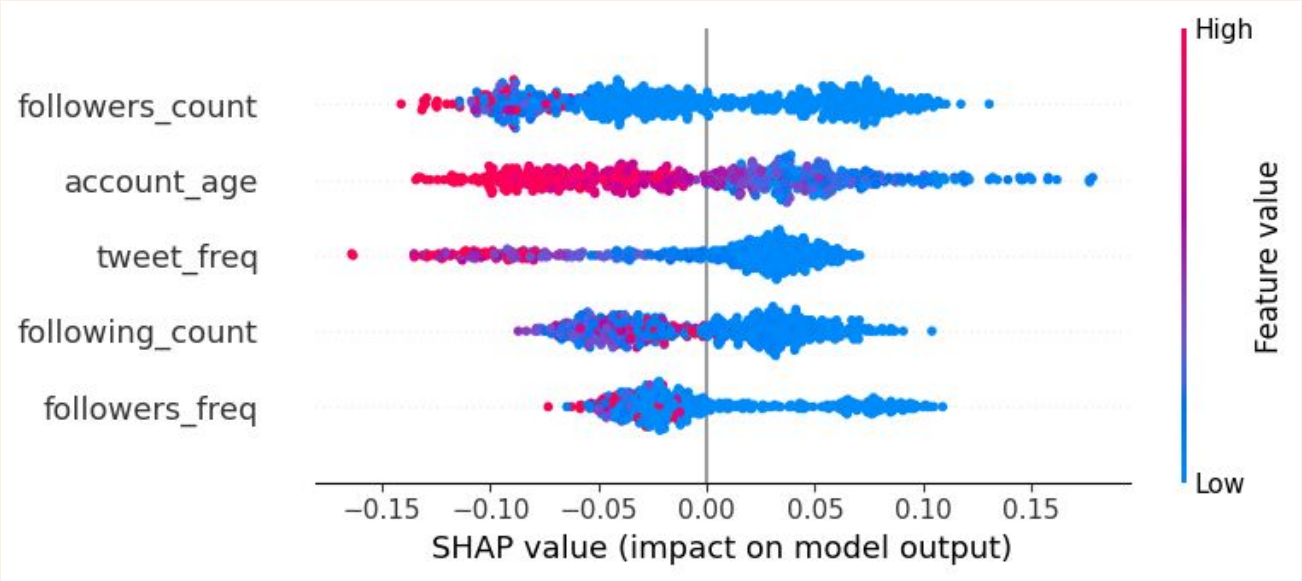
Result

Model Performance Evaluation

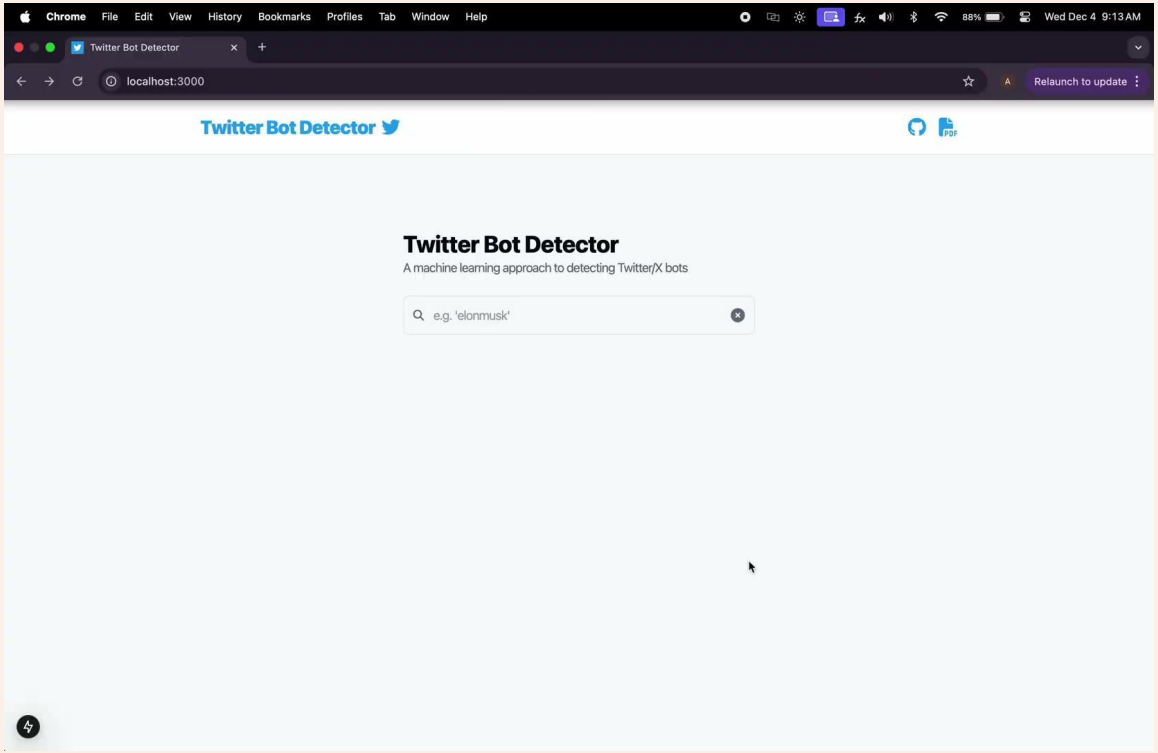
	Precision	Recall	F1-Score
Human	0.8	0.72	0.75
Bot	0.73	0.80	0.76
Weighted Avg.	0.76	0.76	0.76

Result

Model Performance Evaluation



Demo


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Wed Dec 4 9:13 AM
Twitter Bot Detector
localhost:3000
Relaunch to update
Twitter Bot Detector
Twitter Bot Detector
A machine learning approach to detecting Twitter/X bots
e.g. 'elonmusk'

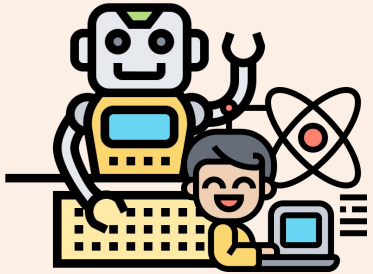
Summary



Select Twitter account to analyze



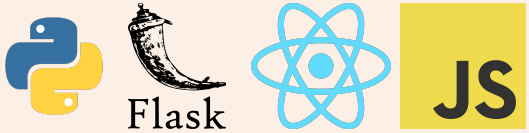
Our app analyzes the account data and uses the model to make a prediction



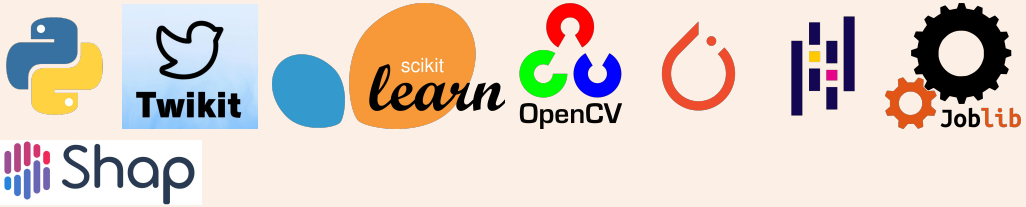
Our app displays a binary result of *human* or *bot*, with standout statistics

Resources Used

Web Application Stack



Machine Learning Stack



References

- Special thanks to the TwiBot-22 team for providing their benchmark dataset
  TwiBot-22: Towards Graph-Based Twitter Bot Detection
  Feng, S., Tan, Z., Wan, H., Wang, N., Chen, Z., Zhang, B., Zheng, Q., et al. (2022).
  arXiv. DOI: 10.48550/arXiv.2206.04564
- Previous Research
  BotArtist: Generic Approach for Bot Detection in Twitter via Semi-automatic Machine Learning Pipeline
  Shevtsov, A., Antonakaki, D., Lamprou, I., Pratikakis, P., Ioannidis, S (2024)
  arXiv. DOI:10.48550/arXiv.2306.00037